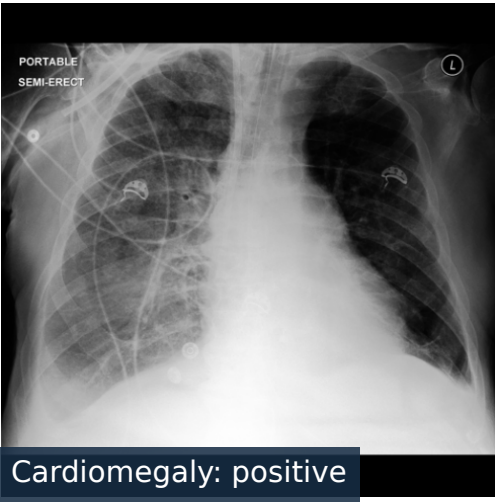


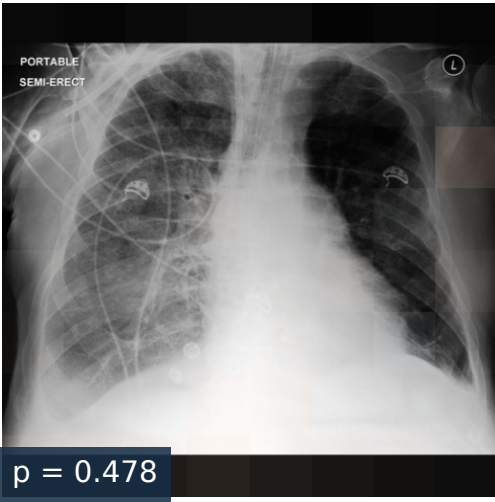
Clinical evidence: a first attribution check

Existing image-only classifiers | Same held-out CXR, two findings | Final 20-epoch heads

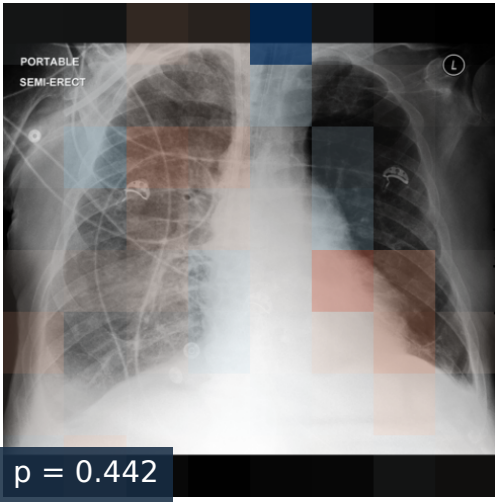
Input + image-level reference



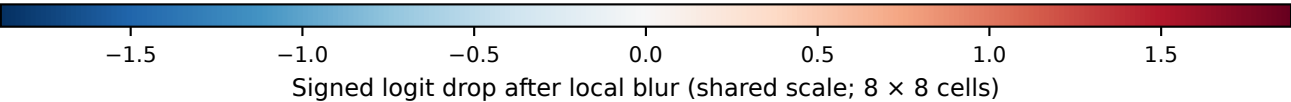
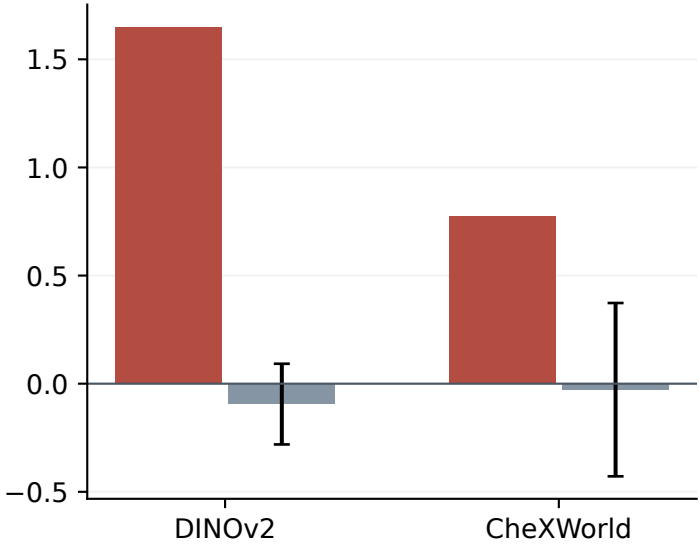
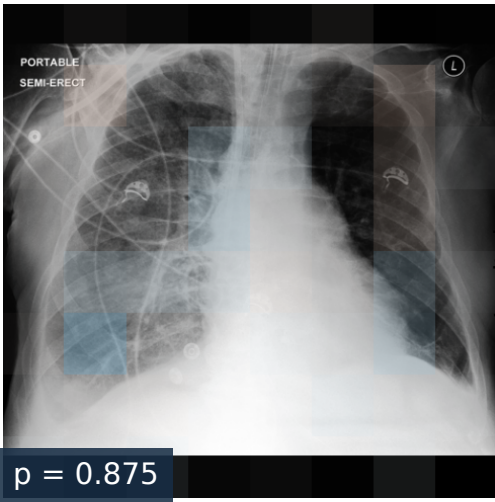
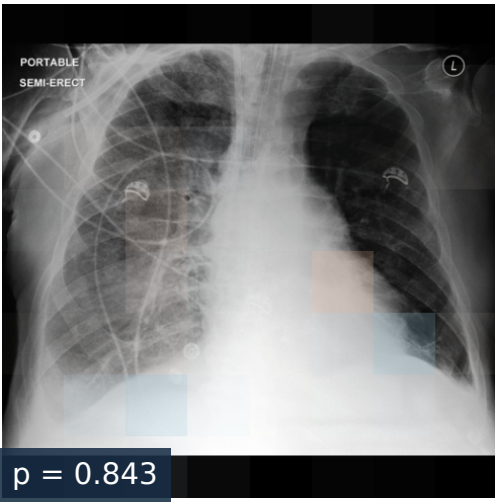
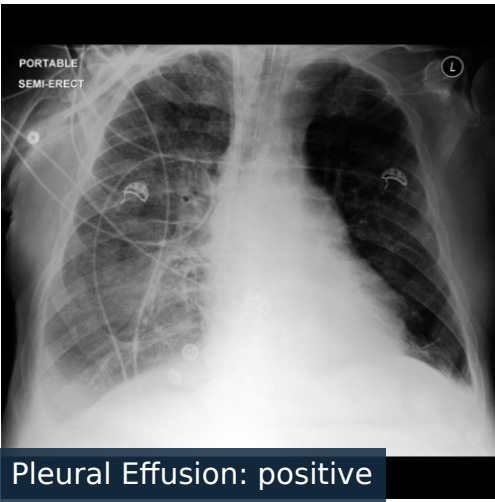
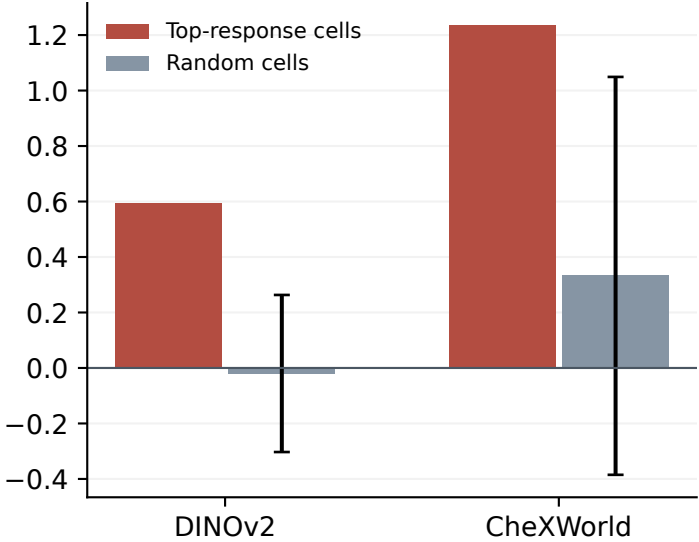
DINOv2-B/14



CheXWorld



Equal input-area blur



Baseline pilot only. No proposed-slot result or lesion-region annotation is available; these maps are attribution, not grounding or attention.

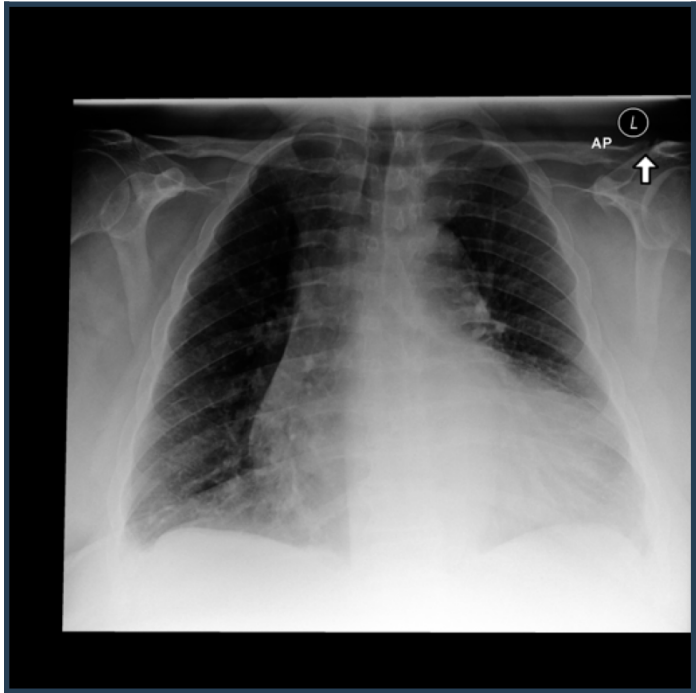
4 / 64 cells blurred per control; random mean ± SD (16 draws).
Equal canvas area; retained image / anatomical area differs.

State-based retrieval: frozen visual-depth pilot

Qwen3.5-0.8B vision tower | image-only input | 447 test images | different-patient retrieval

Fixed query

AP



Recorded positive findings

Cardiomegaly
Pleural effusion

Full-cohort audit

Positive-label Jaccard@3 | 225 queries

Early visual depths **0.173**

Late visual depths **0.217**

All four visual depths **0.193**

Uniform random bank 0.129

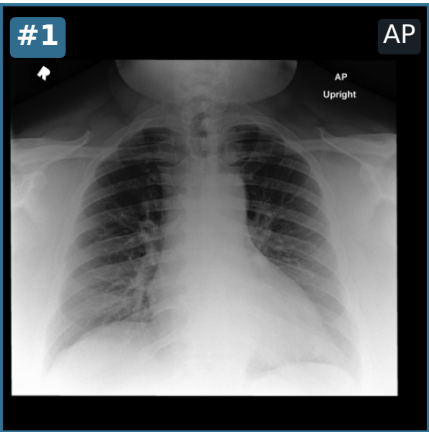
Same-view@3, all 447 queries
90% / 91% / 91% (early / late / all)
Random bank: 57%

Scope

Four frozen vision summaries (slots 5–8).
Fusion slots / full eight-slot state:
not available in this cache.

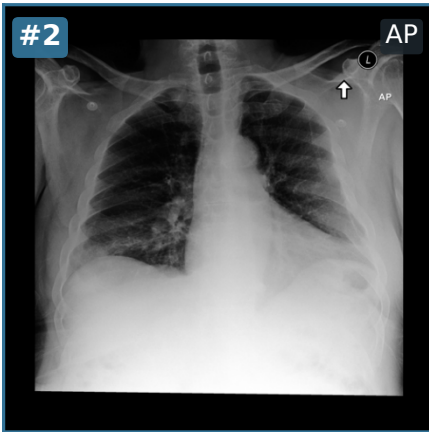
(a) Early visual depths

Blocks 3, 6



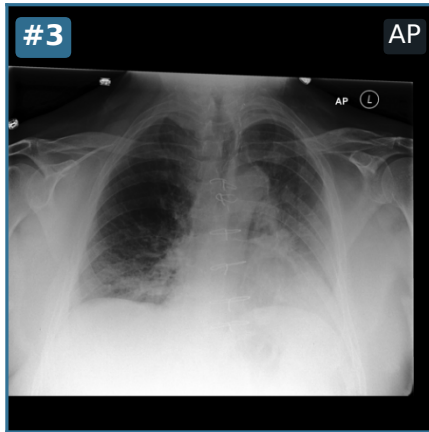
Cosine 0.9931

No recorded positive*



Cosine 0.9921

Atelectasis

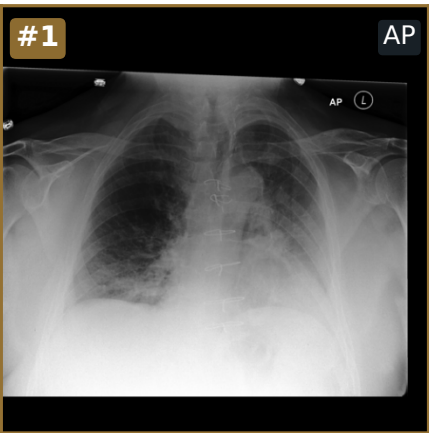


Cosine 0.9920

No recorded positive*

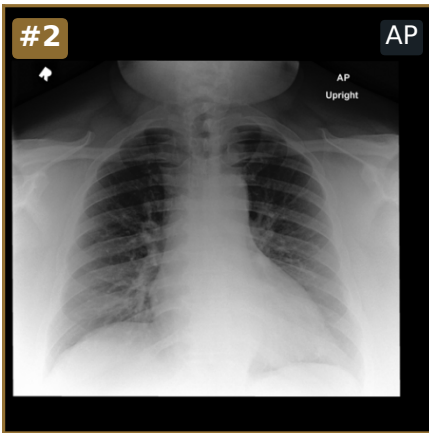
(b) Late visual depths

Blocks 9, 12



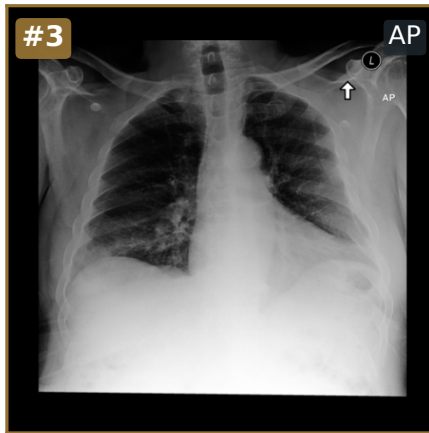
Cosine 0.9921

No recorded positive*



Cosine 0.9919

No recorded positive*

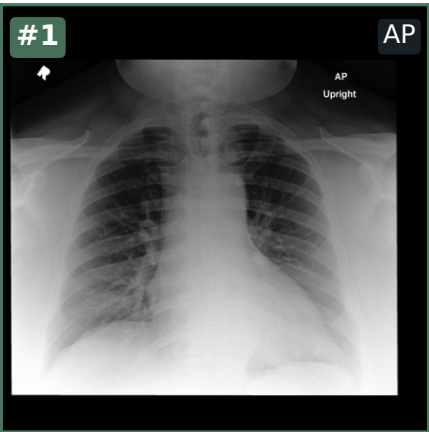


Cosine 0.9917

Atelectasis

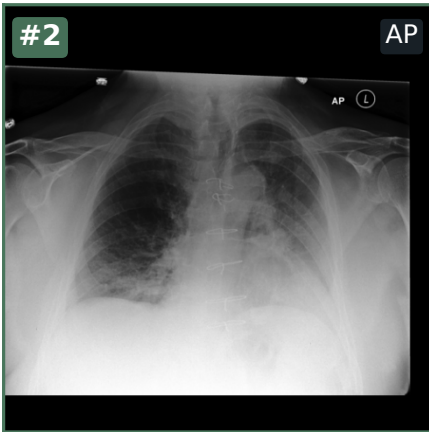
(c) All four visual depths

Blocks 3, 6, 9, 12



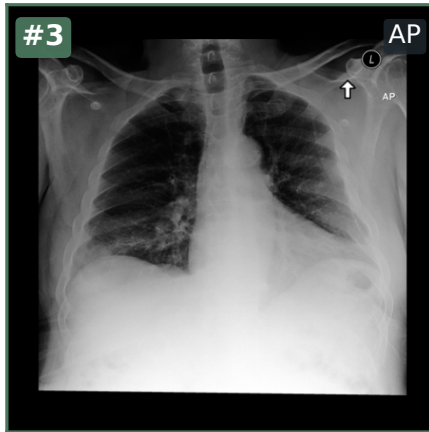
Cosine 0.9925

No recorded positive*



Cosine 0.9920

No recorded positive*



Cosine 0.9919

Atelectasis

*Six prespecified findings; missing / uncertain labels are not negatives. Rank = mean corresponding-depth cosine.

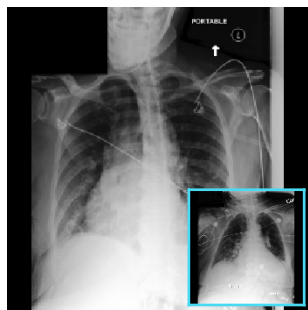
Replacing the slot condition across patients

Same decoder + direct image; only four cached visual slots change

Smallest / largest response in a fixed 16-image test subset; donor shown as inset

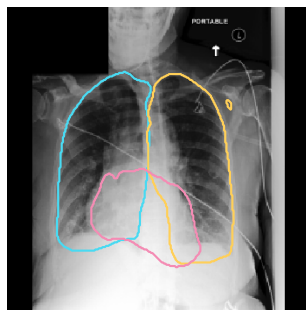
Seg. Smallest response

Fixed image / donor



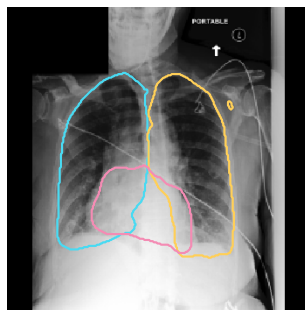
Different-patient donor

Original slots



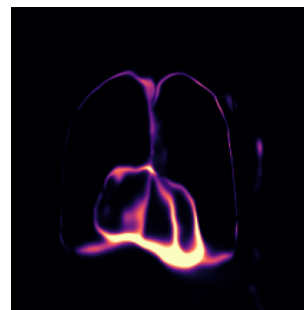
Dice 0.927

Donor slots



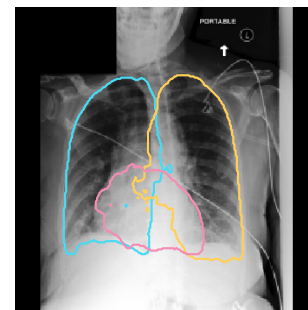
Dice 0.923

Absolute change



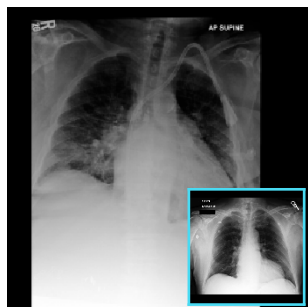
Mean $|\Delta|$ 0.01419

Reference

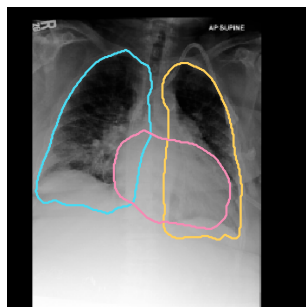


CXAS pseudo-label

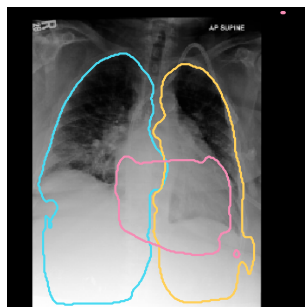
Seg. Largest response



Different-patient donor



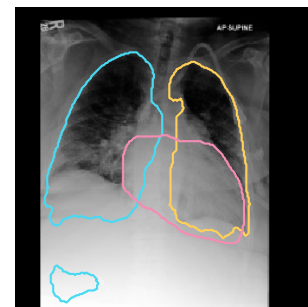
Dice 0.923



Dice 0.746

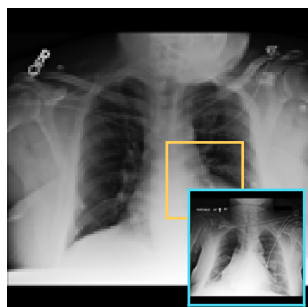


Mean $|\Delta|$ 0.09796

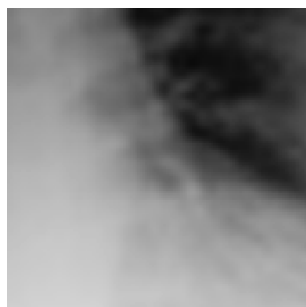


CXAS pseudo-label

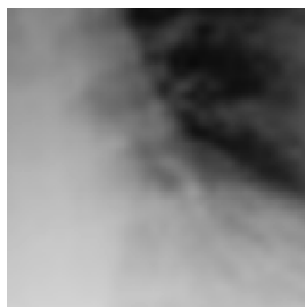
SR Smallest response



Different-patient donor



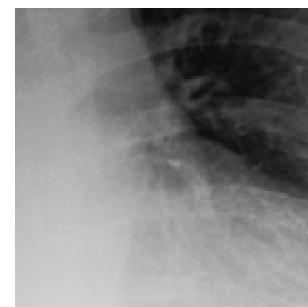
PSNR 32.00 dB



PSNR 31.99 dB

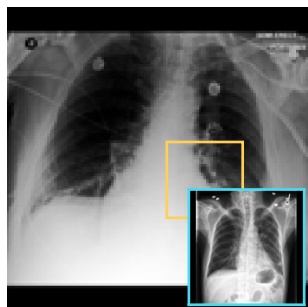


Mean $|\Delta|$ 0.00028



HR crop, reference only

SR Largest response



Different-patient donor



PSNR 30.99 dB



PSNR 30.80 dB

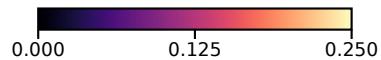


Mean $|\Delta|$ 0.00163

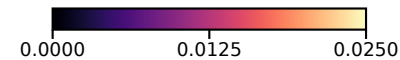


HR crop, reference only

Seg.: mean $|\Delta|$ probability



SR: $|\Delta|$ intensity

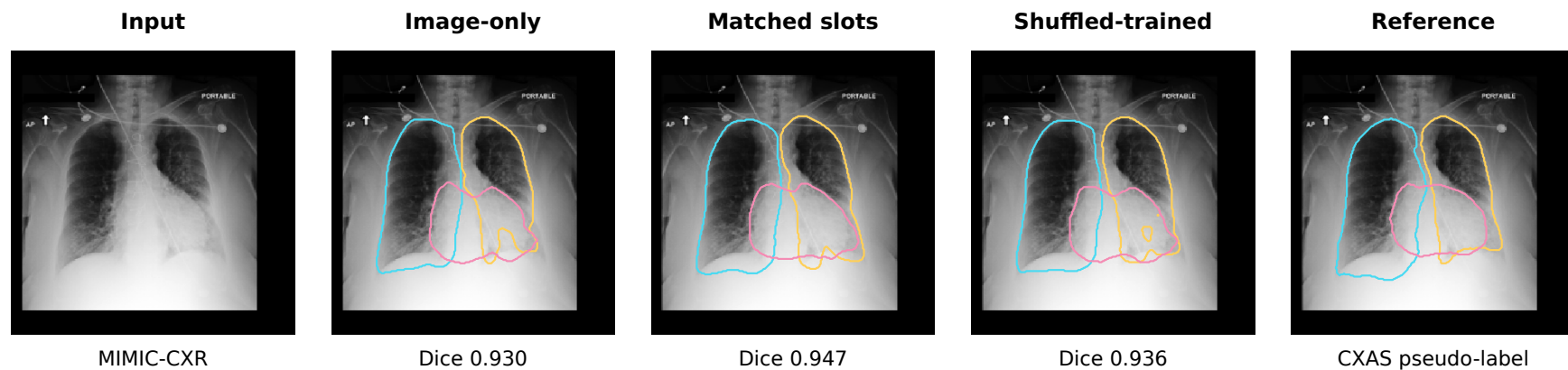


Mean $|\Delta|$ uses the full valid image; SR panels show the same fixed crop. Dark maps preserve weak responses.

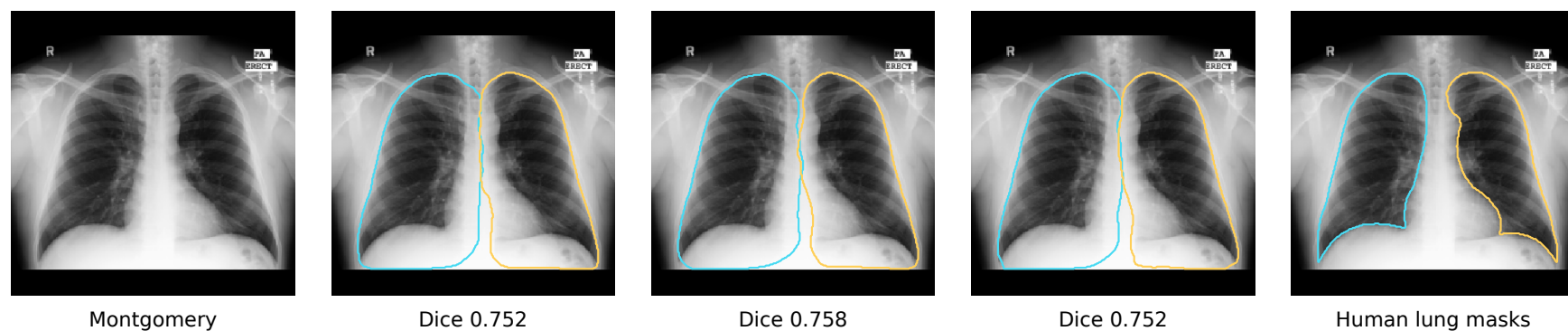
Spatial prediction with frozen visual slots

Qwen3.5-4B visual tower | matched 20-epoch decoders | 4,096 training images

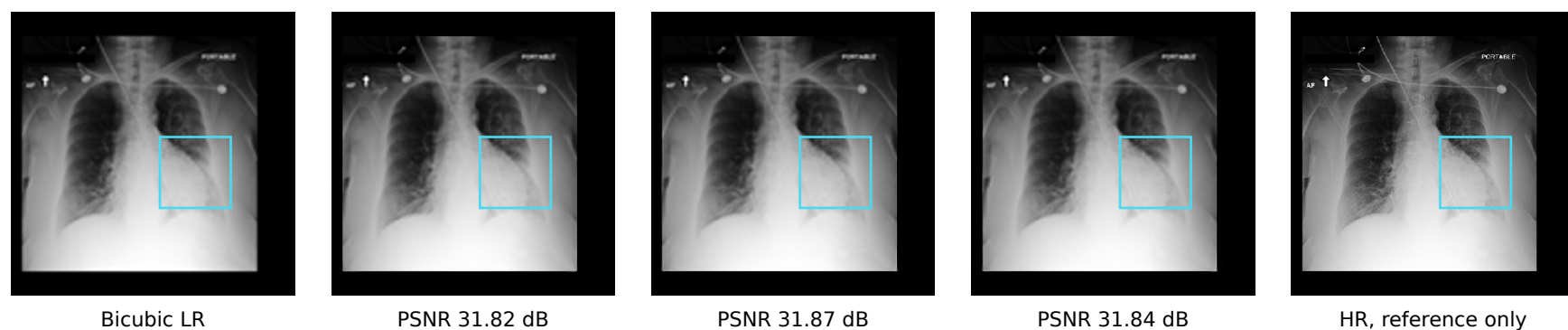
(a) Pseudo reference



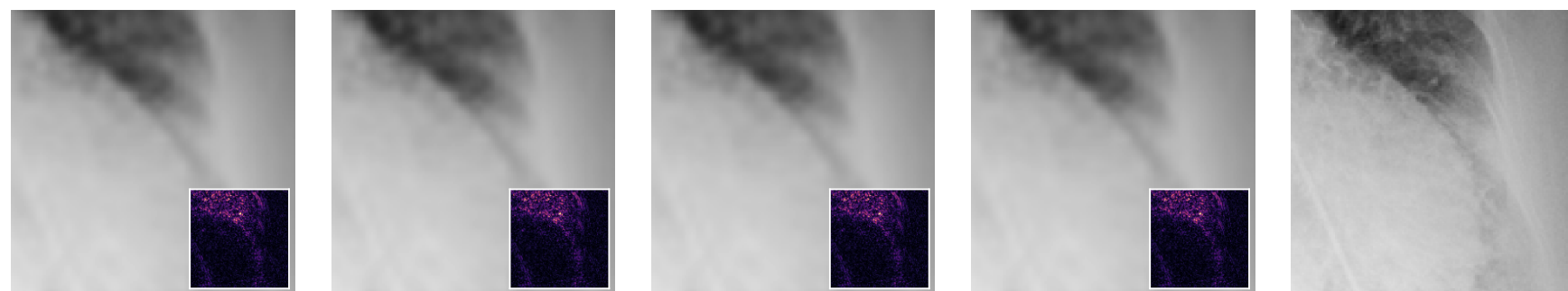
(b) Human reference



(c) 4x super-resolution



(d) Matched 128 px crop



— Right lung — Left lung — Heart (pseudo only)

Inset: absolute HR error

